Edward Kubiak

Full Stack Software Engineer · Agent Systems & Developer Tooling

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Full stack developer, four years building and owning production React applications across Ohio's K-12 education ecosystem. Agents write most of my code now; that moved the work rather than removing it — the judgment, the verification and the consequences stay put, and building the tooling that enforces that is what my open-source work is about. Hired two weeks out of a full-stack certificate in August 2022 to replace an end-of-life AngularJS platform with React, I shipped that rewrite, opened it to districts in September 2024, and remain its sole architect. Creator of **CAST v10.6.1**, a local-first multi-agent control plane for Claude Code whose 46-table execution record is searchable, signed and predictive, shipped as 9 Homebrew packages backed by 4,767 tests; and of **Compute Atlas** (compute-atlas.com), an open, source-cited census of the U.S. grid-scale compute buildout — 2,724 facilities across 50 states, DC and 4 territories, released as open data with a public API.

SKILLS

AI & Agent Systems	Claude Code (hooks, agents, skills), Claude API, CAST v10.6.1 (27 agents, 9 packages), Agent Orchestration & Dispatch, Hook Architecture, MCP (Model Context Protocol), OpenTelemetry (GenAI spans), Deterministic Agent-Reliability Tooling, RAG / Embeddings (FTS5, Ollama)
Languages & Frameworks	TypeScript, JavaScript (Node.js, ESM), Python, Rust, Bash, React 18/19, Next.js 16, Express 4/5, Flask, Vite, Tauri 2, TanStack Query v5 / Table v8, Radix UI, Tailwind CSS v4, Monaco Editor, xterm.js
Data	MS SQL Server, PostgreSQL, SQLite (better-sqlite3, FTS5), BigQuery, Drizzle ORM, SQLAlchemy, Server-Sent Events (SSE)
Testing & Delivery	Vitest, React Testing Library, BATS, pytest / unittest, GitHub Actions, Jenkins, Docker / Docker Compose, Traefik, Homebrew Packaging

PROFESSIONAL EXPERIENCE

Applications Developer · META Solutions — Columbus, OH — Aug 2022 — Present

React 18/19 · Vite 7 · TypeScript · TanStack Query v5 / Table v8 · Radix UI · Monaco Editor · Express 5 · Flask · MS SQL Server · PostgreSQL · Docker · Jenkins · Vitest

- Hired two weeks out of a full-stack certificate to rebuild **CrossCheck** — the EMIS data-validation platform Ohio districts use to catch state-reporting errors before they cost funding — replacing an end-of-life AngularJS 1.x + jQuery + PHP application. A ground-up rewrite across an architecture boundary, not a framework swap: built through year one, internal deployments September 2023, production January 2024, opened to districts September 2024. Sole architect and author of 1,706 of its 1,728 commits as of September 2026.
- Own it end to end: ~54,000 lines, 265 source files, 1,193 tests passing as of August 30, 2026. Led its migration off Create React App to Vite 7, off Bootstrap onto a test-guarded design-token system, and off component-level useEffect fetches onto a typed service and query layer — TanStack Query v5 and Table v8, Radix, Monaco for in-app SQL — one caller at a time, each with a colocated test. TypeScript conversion is incremental.
- Shipped security as artifacts, not assumptions: a CSP pinned by a test that fails when the policy changes, console and debugger stripping on production builds only, a DOMPurify link-hardening hook, and a lint rule failing the build if raw HTML injection appears outside one sanctioned module.
- Built **SES-Wiki** — the Ohio EMIS Student Enrollment Scenario charts district staff consult — from an empty directory in March 2026, then folded it into **CrossCheck** in October 2026: a React 19 + Vite + TypeScript single-page app that builds with **CrossCheck** and is served at /ses/ with no server of its own, signs in through **CrossCheck**'s session, and reads and saves its content through new Flask endpoints in **CrossCheck**'s API — one database row per saved version, ETag revalidation, and admin-only save and import enforced on the server. Running end to end in QA.
- Inherited and modernized the **Customization Web Store** (customizations.metasolutions.net), META's public PowerSchool catalog: Create React App to Vite 7, React 17 to 19, jQuery and Fancybox removed from a live page, an XSS vector closed with DOMPurify sanitization of API-sourced HTML, and an accessibility, SEO and dependency pass documented as seven dated findings. Also maintain the PowerSchool plugin customizations, including the Alert Builder notification system deployed across client districts.
- Own the **E-Rate dashboard** front end on a four-service Docker Compose platform (Flask + SQLAlchemy, PostgreSQL, a Python ingest scraper, Traefik, Jenkins) helping districts capture federal telecom discount funding. Ran an eight-dimension audit with adversarial verification in August 2026 — 98 findings, 3 critical, none refuted — then executed phase one: route guards on six authenticated routes reachable by direct navigation, a hardcoded dev-bypass login removed from the production bundle and grep-verified gone from the shipped artifact, npm audit 387 to 62, and a rebuilt applications table with debounced search, failed-edit rollback and a monotonic sequence race guard.
- Use Claude Code and CAST daily as production infrastructure — the tooling I publish as open source — with the review gate recorded rather than assumed: over the 30 days ending September 4, 2026, **CrossCheck** work passed 147 code-review verdicts, 14 returned with concerns and 2 blocked. It shows in git: 156 commits in all of 2025 against 519 in the first nine months of 2026, and a suite from 884 to 1,193 tests.

OPEN SOURCE — AI DEVELOPER TOOLING

Creator & Maintainer — CAST & Agent-Reliability Tools 2026 — Present

Claude Code · Node.js / ESM · Python · Rust · Bash · SQLite (FTS5) · Tauri 2 · React 19 · Express 5 · OpenTelemetry · BATS · Homebrew

- Creator of **CAST v10.6.1** — a local-first, open-source multi-agent control plane for Claude Code: 27 specialist agents with hook-driven dispatch and model-aware routing, hook-enforced quality gates, and a 46-table SQLite execution record that acts — full-text search over every session (cast ask), signed SHA-256 audit receipts (cast ledger --verify), telemetry-driven cost prediction (cast predict). v10 audited every gate against one question — what does this check report when the thing it guards did not happen? — and rebuilt the ones that reported success either way; new tests are mutation-tested against the bug they guard. 4,767 tests, zero cloud dependencies.
- Shipped it as 9 Homebrew packages plus an umbrella formula: the framework, cast-desktop (native Tauri 2 + React 19 + Rust with an embedded Express 5 + SQLite backend, 11 dashboard views and a PTY-backed terminal), the Claude Code Dashboard v3.0.0 (React 19 + TypeScript + Express 5 + SSE), and standalone packages for agent memory, health checks, journaling, MCP access, signed receipts and dispatch prediction.
- Built deterministic, zero-LLM agent-reliability tools, each shipped with CI — **misfire** and **looptrip** to PyPI and Homebrew, **attest** as a Homebrew formula and Claude Code plugin: **misfire** v0.2.0 audits CLAUDE.md adherence against real traces to find which rules agents actually ignore; **attest** v0.3.0 verifies a subagent's DONE claim against the real git delta; **looptrip** v0.1.2 trips duplicate-work loops at iteration 2 and reproduces prevented spend from a committed fixture.

Creator — Compute Atlas 2026 — Present

Next.js 16 · React 19 · TypeScript · MapLibre GL · Neon Postgres · Drizzle ORM · Vercel

- Built and operate **Compute Atlas** (v1.34.1) — an open, source-cited census of the U.S. grid-scale compute buildout covering data centers, crypto-mining sites and the dedicated power generation contracted to feed them: 2,724 facilities across 50 states, DC and 4 territories, 61.1 GW operational, 129.5 GW under construction and ~491.9 GW planned, every record carrying at least one public source and an explicit confidence level. Built solo on Next.js 16 · React 19 · TypeScript · MapLibre GL · Neon Postgres · Drizzle · Vercel.
- Shipped the whole product: an interactive MapLibre GL map with water and geology overlays, sortable tables, per-facility dossiers, reference hubs (rankings, power, operators, metros, states, AI-vs-crypto), a cited /learn tier and a CORS-open JSON API — the dataset released as open data (code MIT, data CC BY 4.0) and archived on Zenodo (doi:10.5281/zenodo.22284476), now cited by Erin Brockovich's data center reporting map and by Power To The People.
- Engineered an autonomous daily discovery pipeline: a bounded `claude -p` run that discovers, enriches and stages candidates behind a fail-closed kill switch, a heartbeat, extraction guards and a self-reverting cap, verifying every candidate source URL before staging, with human approval required before any record goes live. Extended the survey beyond capacity into its civic footprint — community opposition (litigation, moratoria, referendums, and the projects opposition defeated) and named stakeholders tied to specific sites under a mandatory-citation bar.

AI-ASSISTED ENGINEERING PRACTICE

Condensed; full write-up and dated case studies at edwardkubiak.com/practice.

- Ceremony scales to blast radius, not line count: a one-line edit to an enforcement file gets a written plan; a doc typo gets applied inline.
- The prompt is the specification — if an agent would have to read four files before writing anything, that context belongs in the dispatch, with exact anchors.
- One unit per dispatch, sized to the agent's turn budget: an agent that exhausts it stops mid-sentence with no error, which reads exactly like success.
- Review time goes where it is cheap to be wrong — generated tests get read harder than implementations, because a wrong test is invisible.
- Verify mechanically, then read the verdict: confirm the change is on disk before judging whether it is good. A verdict is evidence about quality, never existence.
- Every new gate is mutation-tested — revert the fix, confirm the check goes red, restore. An assertion that never fails is indistinguishable from one that cannot.

EARLIER CAREER

Nine years of operations and people management preceded the career change — shift and crew leadership, training, performance management, client accounts.

Shift Supervisor · United Parcel Service — Columbus, OH Sep 2020 — Aug 2022

Account & Production Manager · Peabody Landscape Group — Columbus, OH Mar 2013 — Jun 2020

EDUCATION

Full Stack Web Development Certificate — The Ohio State University 2022

Bachelor of Arts and Science in Geological Science — Ohio University 2005–2009